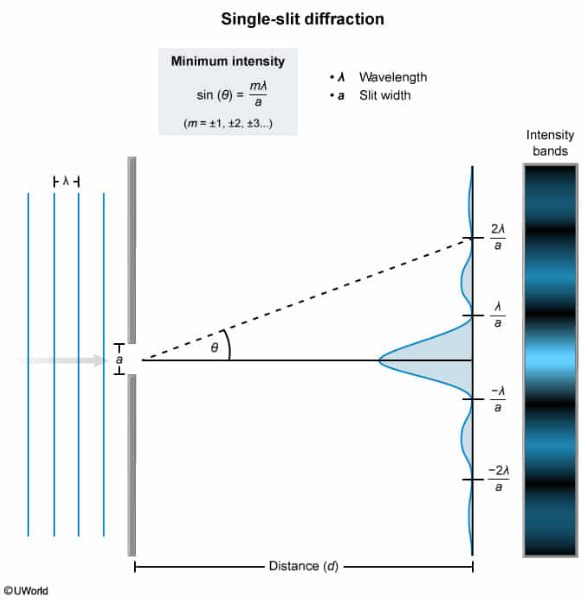
The locations of the bright and dark bands produced by light passing through a single slit depends on the:

I. initial light intensity.
II. slit width.
III. wavelength.

- A. I only
- B. I and II only [10%]
- C. II only [9%]
- D. II and III only [78%]

Correct
79% answered correctly

Explanation:



Diffraction is broadly defined as the bending of light around edges or objects. Classic slit diffraction occurs as a uniform wavefront of monochromatic light arrives at a slit (ie, confined gap) in which the **width of the slit (*a*)** is comparable to the **wavelength of the light (*λ*)**, resulting in the dispersed propagation of light away from the center of the slit.

Diffraction through a single slit produces a pattern of **dark and bright bands** on a flat background surface distant to the slit. Bands occur when the bending of light causes the distance (path length) traveled by some waves to be longer than that of others. When the path lengths of two light waves differ by $\lambda/2$, the waves arrive at the background out-of-phase (ie, the peak of one waveform coincides to a trough of the other). As a result, destructive interference occurs such that dark bands in the diffraction pattern are produced. Conversely, light bands result from zero or minimal destructive interference.

Dark bands (minimum light intensity) due to destructive interference occur when diffracted light waves pass through the slit at angles (θ) that are a function of the slit width and an integer multiple (m = ±1, ±2, ±3…) of the wavelength:

$$\sin\ \theta = \frac{m\,\lambda}{a}$$

According to this relationship, decreased slit width or increased wavelength tends to widen the band pattern such that greater distances are present between light and dark bands. Conversely, increased slit width or decreased wavelength will reduce the distance between the light and dark bands.

Therefore, both **slit width** and **wavelength** influence the angle of propagation associated with the dark and light bands in the pattern of diffracted light **(Numbers II and III)**.

(Number I) The initial intensity of the light affects the intensity of the diffracted light, but does not influence the location of the bright and dark bands in the diffraction pattern.

Educational objective:
Single-slit diffraction is an optical phenomenon that demonstrates the waveform nature of light. The location of the bright and dark bands associated with a given diffraction event is related to slit width and the wavelength of light.

Time spent:0 sec
Subject:
Physics

QID:402147
System:
Light and Sound